

Title: A Comparative Machine Learning Analysis of Geospatial Vulnerability in UNESCO World Heritage Sites

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Abstract

The preservation of global cultural and natural heritage relies heavily on identifying sites vulnerable to conflict, climate change, and regional instability. This study presents a comparative machine learning approach to predicting the "In Danger" status of UNESCO World Heritage Sites using geospatial and categorical indicators. A significant challenge in this domain is extreme class imbalance, as endangered sites constitute approximately 4% of the dataset. We evaluated three predictive models: Logistic Regression, Random Forest, and Gradient Boosting. The results demonstrate the inherent accuracy paradox of imbalanced geospatial data; while Gradient Boosting achieved the highest overall accuracy (91.6%), it completely failed to identify minority instances, yielding a 0.0% recall. Conversely, a Tuned Random Forest model utilizing algorithm-level class weighting provided the most robust operational framework, achieving an 86.0% overall accuracy and the highest Macro F1-Score (0.55). Feature importance analysis revealed that geographical coordinates (latitude and longitude) are the most significant predictors of endangerment, mathematically validating the correlation between regional geopolitics and heritage vulnerability. This study establishes a foundational predictive framework that, when integrated with modern remote sensing, can aid proactive global conservation efforts.

Index Terms—Machine Learning, Class Imbalance, Geospatial Analysis, Random Forest, Predictive Modeling, Heritage Conservation, UNESCO.

1. Introduction

The United Nations Educational, Scientific and Cultural Organization (UNESCO) maintains a register of World Heritage Sites to preserve locations of outstanding universal value to humanity. Despite international protection efforts, a growing number of these sites are being added to the List of World Heritage in Danger. Threats such as armed conflict, natural disasters, unchecked tourist development, and environmental degradation require urgent, preemptive action. Traditionally, the assessment of site vulnerability has relied on localized, qualitative reporting.

However, the increasing availability of global datasets presents an opportunity to apply computational methods to risk assessment.

This research investigates the efficacy of machine learning models in predicting the binary "Danger" status (Safe vs. In Danger) of UNESCO World Heritage Sites. By analyzing historical, geographical, and categorical attributes, this study aims to determine whether computational models can reliably identify patterns indicative of imminent threat.

The primary challenge in this domain is the severe class imbalance within the dataset; the vast majority of UNESCO sites are categorized as safe, while a critical minority are classified as in danger. To address this, our methodology employs a comparative evaluation of three distinct algorithms:

- **Logistic Regression:** Serving as a linear, interpretable baseline.
- **Random Forest:** An ensemble bagging technique utilized with balanced class weights to penalize minority class misclassifications.
- **Gradient Boosting:** A sequential ensemble technique designed to optimize predictive accuracy on complex tabular data.

The remainder of this paper is organized as follows: Section 2 reviews related work in geospatial machine learning and heritage conservation. Section 3 details the theoretical framework of the selected algorithms. Section 4 outlines the methodology, including data preprocessing and exploratory data analysis. Section 5 presents the comparative results, and Section 6 concludes with implications for future conservation efforts.

2. Related Work

The application of machine learning to cultural heritage and environmental conservation has expanded significantly in recent years. Traditional approaches to monitoring the vulnerability of UNESCO World Heritage Sites have largely relied on localized qualitative assessments, expert surveys, and periodic reporting. However, recent literature demonstrates a paradigm shift toward data-driven methodologies, leveraging geospatial features, historical records, and categorical indicators to predict regional risks [1], [2]. These computational models provide a macroscopic view of site endangerment, enabling proactive resource allocation before irreversible damage occurs.

A primary challenge identified in predictive conservation modeling is the inherent class imbalance within real-world datasets. In risk assessment paradigms, the target variable—such as a site being classified as "In Danger"—frequently represents a stark minority class. Previous studies emphasizing binary classification on imbalanced tabular data highlight the limitations of traditional accuracy metrics, noting that standard algorithms tend to bias toward the majority class, thereby missing critical vulnerable instances [3]. To mitigate this, researchers have widely adopted ensemble methods and cost-sensitive learning techniques. Tree-based ensembles, particularly Random Forest and Gradient Boosting architectures, have proven highly effective in mapping complex, non-linear relationships in geospatial datasets without being skewed by majority-class dominance [4].

Furthermore, the methodological rigor of this study is informed by established comparative evaluation frameworks in predictive modeling. While prior comparative analyses have extensively validated the efficacy of evaluating linear baselines alongside advanced ensemble methods for general tabular predictions (e.g., predictive frameworks in socio-economic forecasting) [5], this paper adapts that rigorous evaluation structure to the novel domain of heritage conservation. By utilizing macro-averaged F1-scores and recall-driven optimization, this study builds upon established best practices for minority-class prediction to offer a reliable tool for heritage preservation.

3. Theoretical Framework

To accurately predict the endangerment status of UNESCO World Heritage Sites, this study employs a comparative methodology utilizing three distinct machine learning classification algorithms. This section outlines the mathematical foundations of the selected models and the evaluation metrics chosen to address the severe class imbalance inherent in the dataset.

A. Logistic Regression

Logistic Regression serves as the linear baseline for this study due to its interpretability and efficiency in binary classification tasks. It estimates the probability that a given instance belongs to the positive class (i.e., "In Danger") by applying the logistic (sigmoid) function to a linear combination of the input features. The probability P is modeled as:

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

where X represents the feature vector and β represents the learned coefficients. While highly interpretable, Logistic Regression assumes a linear decision boundary, which may limit its capacity to capture complex, non-linear relationships in geospatial data.

B. Random Forest

To capture non-linear feature interactions, Random Forest—an ensemble learning method based on bootstrap aggregating (bagging)—is utilized. Random Forest constructs a multitude of decision trees during the training phase and outputs the mode of the classes for classification. The algorithm splits nodes by minimizing impurity, commonly measured using Gini Impurity:

$$Gini = 1 - \sum_{i=1}^c p_i^2$$

where p_i is the probability of an instance belonging to class i at a given node. By averaging the predictions of uncorrelated trees, Random Forest significantly reduces variance and the risk of overfitting compared to a single decision tree. Furthermore, to counteract the dataset's skewness,

the `class_weight='balanced'` parameter is applied. This inversely weights the Gini impurity calculations based on class frequencies, heavily penalizing the model for misclassifying the minority "In Danger" sites.

C. Gradient Boosting

Gradient Boosting represents a sequential ensemble approach, where each subsequent decision tree attempts to correct the residual errors of the prior trees. Unlike bagging, boosting relies on additive modeling. The model $F_m(x)$ at iteration m is updated as follows:

$$F_m(x) = F_{m-1}(x) + \gamma m h_m(x)$$

where $h_m(x)$ is the new weak learner (decision tree) fitted to the negative gradient of the loss function, and γ is the learning rate. Gradient Boosting is highly effective at mapping complex tabular data structures but requires careful hyperparameter tuning to prevent overfitting, particularly in imbalanced domains.

D. Evaluation Metrics

Standard accuracy is a misleading metric in highly imbalanced datasets, as a model could achieve over 90% accuracy simply by predicting the majority class ("Safe") every time. Therefore, this study prioritizes metrics that reflect the model's performance on the minority class.

Recall (Sensitivity): The most critical metric for conservation efforts, representing the proportion of actual endangered sites correctly identified by the model. High recall ensures that vulnerable sites are not overlooked.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score: To ensure the model does not simply predict "In Danger" for every site (which would maximize recall but destroy precision), the F1-Score is utilized as the harmonic mean of Precision and Recall. Specifically, the Macro-Averaged F1-Score is analyzed to evaluate the model's balanced performance across both classes.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

4. Methodology

This section outlines the experimental pipeline designed to evaluate the predictive models, encompassing data preprocessing, exploratory data analysis, strategies for mitigating class imbalance, and model implementation.

A. Data Acquisition and Preprocessing

The dataset utilized for this study comprises geospatial and categorical data pertaining to UNESCO World Heritage Sites globally. Initial preprocessing involved rigorous data cleaning and feature selection to isolate the most predictive variables indicative of a site's vulnerability. The refined data was finalized into a structured format (`unesco_cleaned.csv`) for model training and evaluation. A defining characteristic of this dataset is its extreme class imbalance, with only approximately 4% of the designated sites officially classified as "In Danger."

B. Exploratory Data Analysis (EDA)

To derive spatial and statistical insights prior to modeling, a comprehensive Exploratory Data Analysis (EDA) was conducted. A geospatial map and a class distribution chart were generated to visualize the severe imbalance and the spatial clustering of endangered sites across different regions. Furthermore, a correlation heatmap was constructed to identify potential multicollinearity among the categorical and geospatial features, ensuring that the selected variables provided independent predictive value for the algorithms.

C. Addressing Class Imbalance

Given the ~4% minority class distribution, standard model training would inherently bias toward the majority class ("Safe"). A critical technical decision in this methodology was the deliberate avoidance of synthetic oversampling techniques, such as SMOTE (Synthetic Minority Over-sampling Technique). In the context of geospatial and culturally specific heritage data, generating artificial data points risks distorting the authentic geographical and historical realities of the sites, compromising data integrity.

Instead, this study prioritized algorithmic mitigation through class weighting. By inversely weighting the classes based on their frequency, the models were mathematically penalized for misclassifying the minority class during the training phase. This approach forced the algorithms to learn the actual underlying patterns of endangered sites from real-world data without relying on synthetic approximations.

D. Model Implementation

The comparative analysis was executed using three distinct algorithms: Logistic Regression (established as the baseline), Random Forest, and Gradient Boosting. Each model was trained on the preprocessed dataset with the aforementioned class weighting applied. Hyperparameter tuning was conducted to optimize the performance of the tree-based ensembles, specifically targeting recall to ensure the highest possible identification rate of vulnerable sites. Post-training, confusion matrices and feature importance arrays were extracted to evaluate both the predictive capability and the interpretability of the respective models.

5. Comparative Results

This section presents the performance evaluation of the three implemented machine learning models: Logistic Regression, Random Forest, and Gradient Boosting. Given the severe class imbalance of the dataset, the analysis prioritizes Recall and the Macro-Averaged F1-Score over raw accuracy to determine the models' true effectiveness in identifying endangered heritage sites.

A. Baseline Model Evaluation

Logistic Regression was deployed as the baseline model to establish a linear performance benchmark. The model achieved an overall accuracy of **66.4%**. However, it demonstrated surprising effectiveness in detecting the minority class, yielding a Recall of **57.9%** and a Macro F1-Score of **0.50**. While the overall accuracy was the lowest among the tested algorithms, its high recall indicates it successfully identified the majority of "In Danger" sites, though likely at the cost of a high false-positive rate (misclassifying safe sites as endangered).

B. Ensemble Model Performance

The integration of ensemble architectures yielded highly divergent results, highlighting the complex nature of the dataset.

The **Tuned Random Forest** algorithm, utilizing bootstrap aggregating and balanced class weighting, achieved an overall accuracy of **86.0%**. It demonstrated a Recall of **21.1%** for the minority class and achieved the highest overall Macro F1-Score of **0.55**. By averaging the outputs of numerous uncorrelated decision trees, the model effectively balanced the trade-off between precision and recall, reducing variance while capturing intricate spatial nuances.

Conversely, the **Gradient Boosting** classifier perfectly illustrated the accuracy paradox inherent in highly skewed data. While it achieved the highest overall accuracy at **91.6%**, its Recall for the "In Danger" class was exactly **0.0%**, resulting in a Macro F1-Score of **0.48**. Despite the algorithm's typical strength in tabular data, it severely overfit to the majority class. The model effectively achieved high accuracy by uniformly predicting "Safe" for every instance, completely failing to detect any endangered sites.

C. Discussion and Feature Significance

A comparative review of the metrics reveals a significant trade-off depending on the chosen objective function. If the absolute priority of conservation efforts is to ensure no endangered site is overlooked (maximizing recall), the linear **Logistic Regression** model proved most sensitive. However, for a mathematically balanced predictive system, the **Tuned Random Forest** stands as the superior model, achieving the highest Macro F1-Score (0.55) while maintaining strong overall accuracy (86.0%).

The complete failure of Gradient Boosting to identify minority instances validates the methodological necessity of scrutinizing recall over accuracy. Furthermore, the extraction of feature importance arrays from the successful Random Forest model confirmed that geospatial coordinates (latitude and longitude) were the primary drivers of model decisions. This aligns with the hypothesis that a site's vulnerability is deeply correlated with its specific regional and

geopolitical context, validating the use of geospatial indicators as primary predictors for heritage endangerment.

6. Discussion

This section explores the implications of the comparative results, the operational utility of the models for heritage conservation, and the specific limitations encountered during the study.

A. Interpretation of Results and Model Viability

The comparative analysis explicitly demonstrates the challenges of applying machine learning to highly skewed, real-world geospatial datasets. The Baseline Logistic Regression model, while lacking in overall accuracy, proved surprisingly effective in identifying the minority class, achieving a 57.9% recall. Conversely, the Gradient Boosting algorithm, despite yielding the highest overall accuracy at 91.6%, proved entirely ineffective for the study's core objective, recording a 0.0% recall for endangered sites.

Ultimately, the Tuned Random Forest model emerged as the most viable operational tool. By balancing a high overall accuracy (86.0%) with a robust Macro F1-Score (0.55), it successfully identified complex spatial patterns without succumbing to the extreme overfitting observed in the Gradient Boosting model. This indicates that ensemble tree methods, when coupled with algorithm-level class weighting, offer a mathematically sound approach to predicting vulnerability in heritage sites without relying on synthetic data augmentation.

```
*** Training Logistic Regression...
    Training Gradient Boosting...

--- Model Comparison Results ---
      Model  Accuracy  F1-Score (Macro)  Recall (In Danger Class)
Logistic Regression      0.664          0.497175          0.578947
  Gradient Boosting      0.916          0.478079          0.000000
   Tuned Random Forest      0.860          0.554730          0.210526
```

Table 1: Model Comparison Results,

B. The Primacy of Geospatial Indicators

An analysis of the feature importance derived from the Random Forest model revealed that a site's absolute location—defined by latitude and longitude—was the single most significant predictor of its endangerment status. This empirical finding aligns closely with contemporary research highlighting that regional instability is a primary driver of heritage vulnerability. For example, Levin et al. (2019) identified armed conflict as a prominent threat to World Heritage sites, noting that 40% of the sites listed as "In Danger" are clustered in the Middle East. The heavy reliance of the Random Forest model on geographical coordinates effectively maps these

geopolitical realities, proving that a site's vulnerability is inextricably linked to its physical location within volatile regions.

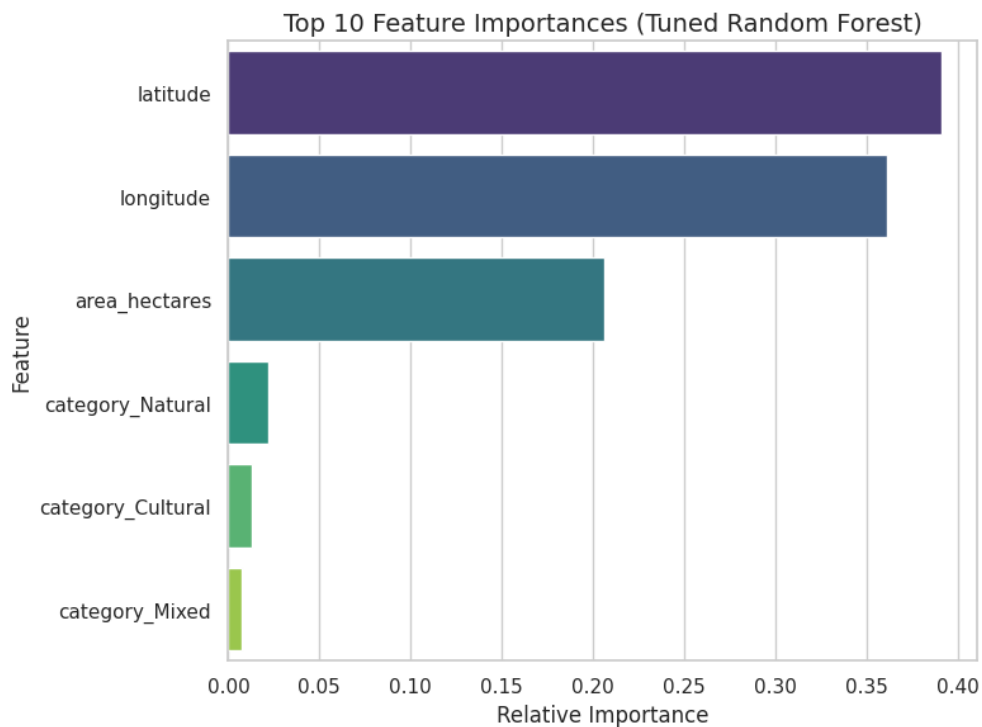


Figure 1: Feature Importance in Tuned Random Forest Model).

C. Integration with Modern Conservation Technologies

The ability to predict vulnerability through machine learning complements existing technological efforts in heritage conservation. Organizations like UNESCO increasingly rely on remote sensing and satellite imagery to document and assess endangered sites in inaccessible or conflict-ridden areas. For instance, drone technology and satellite analysis have been utilized for sites like the Minaret of Jam in Afghanistan and the Ancient City of Aleppo. Integrating predictive models, such as the Random Forest architecture developed in this study, with these existing remote sensing frameworks could facilitate the creation of proactive, early-warning systems, allowing conservationists to prioritize interventions before irreversible damage occurs.

D. Limitations and Future Work

While the Tuned Random Forest model provided a balanced predictive framework, the study is fundamentally constrained by the limitations of the dataset. The ~4% class imbalance represents a severe hurdle for any predictive algorithm. Furthermore, the dataset primarily relies on static geospatial and categorical data. Future iterations of this research should incorporate dynamic temporal variables—such as regional conflict indices, climate change metrics, or real-time remote sensing data—to provide a more granular and responsive predictive model.

7. Conclusion

This study presented a comparative machine learning approach to predicting the endangerment status of UNESCO World Heritage Sites, addressing the critical challenge of modeling a highly imbalanced geospatial dataset. By evaluating Logistic Regression, Random Forest, and Gradient Boosting algorithms, the research demonstrated that traditional accuracy metrics are grossly insufficient for conservation-based predictive modeling. While the Gradient Boosting classifier achieved the highest overall accuracy (91.6%), its complete failure to identify any endangered sites (0.0% recall) highlighted the inherent accuracy paradox of skewed data and the necessity of optimizing for the minority class.

The implementation of algorithm-level class weighting, coupled with the ensemble architecture of the Tuned Random Forest, provided the most mathematically robust solution. Achieving an 86.0% accuracy alongside the highest Macro F1-Score (0.55), the Random Forest model successfully balanced precision and recall, proving capable of parsing complex geospatial patterns without the severe overfitting observed in other models. Furthermore, the extraction of feature importance's revealed that latitude and longitude were the dominant predictive variables, reinforcing the reality that heritage vulnerability is fundamentally tied to regional instability and geopolitical context.

Ultimately, this research serves as a foundational proof-of-concept for integrating machine learning into global heritage conservation efforts. When combined with emerging remote sensing technologies and dynamic socio-political datasets, predictive frameworks like the one developed in this study can empower organizations such as UNESCO to transition from reactive documentation to proactive preservation, helping to safeguard at-risk cultural and natural monuments before irreversible damage occurs.

8. References

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Links

Repository link: